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VEHICLE UPPER STRUCTURE

Abstract

The vehicle upper structure has a hatchback door and a rail garnish covering the roof side rail. The rail garnish is provided along the roof side rail and includes a base whose rear end extends to near the front end of the hatchback door (upper end) in the closed state, and a wall portion that extends downward from the rear end of the base. The upper end of the rear side surface of the wall portion is sloped backward.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-026927 filed on Feb. 26, 2024, which is incorporated herein by reference in its entirety including the specification, claims, drawings, and abstract.

TECHNICAL FIELD

[0002] The present disclosure relates to improvements to vehicle upper structure.

BACKGROUND

[0003] Conventionally, vehicles are equipped with rail garnishes covering roof side rails (e.g., Patent Document 1). Roof side rails are frame members that connect the upper ends of the front pillar, center pillar, and rear pillar and extend in the front-rear direction of the vehicle. The rail garnish is primarily a decorative component.

CITATION LIST

[0004] PATENT LITERATURE 1: JP2019-026239A

[0005] FIG. **2** is a side view of a vehicle with a hatchback door **10**, which is a flip-up type door at the rear of the vehicle, and a rail garnish **14** covering the roof side rail **12**. FIG. **3** is an enlarged cross-sectional view of part A of FIG. **2** and a cross-sectional view of a conventional vehicle upper structure **20**. In the drawings to which this specification refers, "FR" represents forward in the front-back direction of the vehicle, and the opposite direction of "FR" represents backward in the front-back direction of the vehicle. In this specification, forward in the vehicle front-back direction is simply described as forward, and backward in the vehicle front-back direction is simply described as backward. UP″ represents vertical upward, and the opposite direction of 'UP' represents vertical downward. In this specification, when a position is described as above a certain location, it means not only vertically upward from that location, but also diagonally upward from that location. When the term "below a certain position" is used herein, it means not only vertically downward from the said position but also diagonally downward from the said position.

[0006] The rail garnish **14** that the conventional vehicle upper structure **20** has is described as rail garnish **14***a*. The rail garnish **14***a* is formed of steel plate. Referring to FIG. **3**, the rail garnish **14***a* is provided along the roof side rail **12** and comprises a base **22** extending in the front-rear direction of the vehicle and a wall portion **24** extending downwardly from the rear end **22***a* of the base **22**. In particular, the wall portion **24** of the rail garnish **14***a* extends straight and almost vertically downward from the rear end **22***a* of the base **22**.

[0007] The rear end **22***a* of the base **22** extends to the vicinity of the front end **10***a* of the hatchback door **10** in the closed state. The sign **10***b* shown in FIG. **3** is the upper end of the hatchback door **10** in the closed state. As shown by the solid line in FIG. **3**, in the closed state, the upper end **10***b* extends in the front-rear direction of the vehicle (in an abbreviated horizontal direction). The front end **10***a* of the hatchback door **10** is the front end of the upper end **10***b*.

[0008] The hatchback door **10** opens and closes by rotating around a rotation axis parallel to the vehicle width direction. When the hatchback door **10** opens from the closed state, the upper end **10***b* rotates so that the front end **10***a* moves downward, as indicated by the arrow AR in FIG. **3**. In FIG. **3**, the rotation of the upper end **10***b* is shown as a dashed line.

[0009] When the rotation axis of the hatchback door **10** is lower than the upper end **10***b* of the hatchback door **10** in the closed state (not to mention that the rotation axis is behind the front end **10***a* of the hatchback door **10**), as the hatchback door **10** (upper end **10***b*) rotates from the closed state to the open state, the front end **10***a* moves forward. Here, since the rotating upper end **10***b* must not contact the wall portion **24** of the rail garnish **14***a*, the wall portion **24** is provided in front of the most front position Pf in the movement path of the front end **10***a* as the hatchback door **10** opens and closes (that is, the rotation of the upper end **10***b*).

[0010] Thus, in the past, the wall portion **24** was provided almost vertically in front of the most front position Pf. The front end **10***a* of the hatchback door **10** in the closed state may be located

rearward of the most front position Pf. In such a case, the gap G**1** between the rail garnish **14***a* (rear end **22***a* of the base **22**) and the closed hatchback door **10** (front end **10***a* of the upper end **10***b*) is inevitably larger than the gap between the wall portion **24** and the most front position Pf. In other words, the problem arose that the sight gap G**1** became larger.
[0011] The purpose of the vehicle upper structure disclosed herein is to reduce the sight gap between the rail garnish and the hatchback door.
SUMMARY
[0012] A vehicle upper structure disclosed in the present specification comprising: [0013] a hatchback door; and [0014] a rail garnish covering a roof side rail, wherein [0015] the rail garnish includes: [0016] a base provided along the roof side rail, and having a rear end which extends to a region proximate a front end of the hatchback door in a closed state; and [0017] a wall extending downward from a rear end of the base, and [0018] an upper end portion of a rear side surface of the wall is inclined forward in a front-rear direction of the vehicle toward a lower side.
[0019] A lower end portion of the rear side surface of the wall may be inclined rearward in the front-rear direction of the vehicle toward a lower side.
[0020] A position of the lower end portion of the rear side surface of the wall in the front-rear direction of the vehicle may be at the same position as a position of the upper end portion of the rear side surface of the wall in the front-rear direction of the vehicle, or is at a rear side in the front-rear direction of the vehicle than the position of the upper end portion of the rear side surface of the wall in the front-rear direction of the vehicle.
[0021] The rear side surface of the wall may be curved along a trajectory of movement of the front end of the hatchback door due to opening or closing of the hatchback door.
[0022] The rail garnish may be made of a resin.
[0023] The vehicle upper structure disclosed herein allows for a smaller sight gap between) the rail garnish and the hatchback door.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0024] Embodiment of the present disclosure will be described based on the following figures, wherein:
[0025] FIG. **1** is cross-sectional view of the vehicle upper structure.
[0026] FIG. **2** is side view of a vehicle with a hatchback door and rail garnish.
[0027] FIG. **3** is cross-sectional view of a conventional vehicle upper structure.
EMBODIMENTS
[0028] FIG. **1** is a cross-sectional view of the **30** vehicle upper structure. FIG. **1** is an enlarged cross-sectional view of part A of FIG. **2**, as well as FIG. **3**. However, the shape of the rail garnish **14** differs from each other between the vehicle upper structure **30** of this embodiment and the conventional vehicle upper structure **20**. The rail garnish **14** of the vehicle upper structure **30** of this embodiment is described as rail garnish **14***b*.
[0029] The vehicle upper structure **30** has a hatchback door **10** and a rail garnish **14***b* covering the roof side rail **12**. The structure of the hatchback door **10** and its movement when opening and closing in the vehicle upper structure **30** are the same as those of the conventional vehicle upper structure **20**, so the same symbols as in FIG. **3** are used and their description is omitted.
[0030] The rail garnish **14***b* is made of resin, unlike the conventional rail garnish **14***a*. The rail garnish **14***b* comprises a base **32** provided along the roof side rail **12** and a wall portion **34** extending downward from the rear end **32***a* of the base **32**. The base **32**, like the base **22** of the conventional rail garnish **14***a*, extends in the front-rear direction of the vehicle and its rear end **32***a* extends to the vicinity of the front end **10***a* of the hatchback door **10** (upper end **10***b*) in the closed

state.

[0031] In the rail garnish **14***b*, the upper end of the rear side surface **34***a* of the wall portion **34** is sloped downward and forward. In other words, the upper end of the rear side surface **34***a* is a rearward and downward facing slope. In this embodiment, like the rear side surface **34***a*, the upper end of the front side surface **34***b* of the wall portion **34** also slopes forward. In other words, in this embodiment, the upper end of the wall portion **34** is warped downward and forward. However, it is sufficient that at least the upper end of the rear side surface **34***a* slopes forward, and the upper end of the front side surface **34***b* need not necessarily slope forward. For example, the front side surface **34***b* may be a plane parallel to the vertical direction.

[0032] The lower end of the rear side surface **34***a* may be sloped backward toward the bottom. In other words, the lower end of the rear side surface **34***a* is a slope facing backward and upward. In this embodiment, like the rear side surface **34***a*, the lower end of the front side surface **34***b* also slopes backward. In other words, in this embodiment, the lower end of the wall portion **34** is warped backward toward the bottom. However, it is sufficient that at least the lower end of the rear side surface **34***a* slopes backward, and the lower end of the front side surface **34***b* need not necessarily slope backward. For example, as described above, the front side surface **34***b* may be a plane parallel to the vertical direction.

[0033] The position P**1** of the lower end of the rear side surface **34***a* in the longitudinal direction of the vehicle may be the same position P**2** of the upper end of the rear side surface **34***a* in the longitudinal direction of the vehicle. Alternatively, as shown in FIG. **1**, the position P**1** of the lower end of the rear side surface **34***a* in the vehicle longitudinal direction may be rearward of the position P**2** of the upper end of the rear side surface **34***a* in the vehicle longitudinal direction.

[0034] In this embodiment, the rear side surface **34***a* is curved along the movement path (see arrow AR in FIG. **1**) of the front end **10***a* of the hatchback door **10** (upper end **10***b*) as the hatchback door **10** is opened and closed. In other words, the rear side surface **34***a* has a circular arc shape in the side view. In other words, as the rear side surface **34***a* moves downward from the upper end of the rear side surface **34***a* (position P**2**), the position of the rear side surface **34***a* gradually moves forward in the front-back direction of the vehicle. Then, position P**3**, which is opposite to (at the same height as) the most front position Pf in the movement trajectory of the front end **10***a*, becomes the most front position in the rear side surface **34***a*. Position P**3** is the same position as the position of the rear side of the wall portion **24** in the front-back direction of the vehicle in the conventional vehicle upper structure **20** (see FIG. **3**). Furthermore, as one moves downward from position P**3**, the position of the rear side surface **34***a* in the front-back direction of the vehicle gradually moves backward to the lower end (P**1**).

[0035] The shape of the rear side surface **34***a* of the vehicle upper structure **30** of this embodiment, in particular, the wall portion **34** of the rail garnish **14***b*, is as described above. In this embodiment of the vehicle upper structure **30**, the upper end of the rear side surface **34***a* of the wall portion **34** is sloped downward and forward, so that, at least compared to the vehicle upper structure **20** as shown in FIG. **3** (in which the wall portion **24** (rear side) extends straight and almost vertically downward), the rail garnish **14** G**2** between the rail garnish **14***b* and the hatchback door **10** in the closed state can be reduced. Specifically, the clearance gap G**2** of the vehicle upper structure **30** is smaller than the difference between position P**3** and position P**2** in the front-rear direction of the vehicle and the clearance gap G**1** of the vehicle upper structure **20**.

[0036] The lower end of the rear side surface **34***a* of the wall portion **34** is sloped downward and backward, so that even when the hatchback door **10** is opened (in other words, when the front end **10***a* of the hatchback door **10** is moved backward), the clearance gap between the rail garnish **14***b* and the hatchback door **10** The gap between the rail garnish **14***b* and the hatchback door **10** can be reduced. In addition, the position P**1** of the lower edge of the rear side surface **34***a* in the longitudinal direction of the vehicle is the same as or further rearward than the position P**2** of the upper edge of the rear side surface **34***a*, so that even when the hatchback door **10** is opened wider so that the front end **10**_a_ of the hatchback door **10** is the same as or further rearward than when the door is closed The gap between the rail garnish **14**_b_ and the hatchback door **10** can be reduced.

[0037] In addition, as in the present embodiment, the rear side surface **34**_a_ is curved along the path of movement of the front end **10**_a_ as the hatchback door **10** opens and closes, so that the clearance between the rail garnish **14**_b_ and the hatchback door **10** can be kept small and constant regardless of how the hatchback door **10** opens. The curved track of the hatchback door **10** allows the gap between the rail garnish **14**_b_ and the hatchback door **10** to be kept small and constant regardless of how the hatchback door **10** opens.

[0038] Furthermore, the fact that the rail garnish **14**_b_ is made of resin makes its molding easier than when the rail garnish **14**_b_ is at least made of steel plate. Therefore, by having the rail garnish **14**_b_ made of resin, the shape of the wall portion **34** (especially the rear side surface **34**_a_) can be easily molded as described above.

[0039] Although the embodiments of the vehicle upper structure of the present disclosure have been described above, the vehicle upper structure of the present disclosure is not limited to the above embodiments, and various modifications are possible without departing from the intent thereof.

REFERENCE SIGNS LIST

[0040] **10** hatchback door, **10**_a_ front end, **10**_b_ upper end, **12** roof side rail, **14**, **14**_a_, **14**_b_ rail garnish, **20**,**30** vehicle upper structure, **22**,**32** base, **22**_a_,**32**_a_, rear end, **24**,**34**, wall portion, **34**_a_, rear side surface, **34**_b_, front side surface.

Claims

1. A vehicle upper structure comprising: a hatchback door; and a rail garnish covering a roof side rail, wherein the rail garnish includes: a base provided along the roof side rail, and having a rear end which extends to a region proximate a front end of the hatchback door in a closed state; and a wall extending downward from a rear end of the base, and an upper end portion of a rear side surface of the wall is inclined forward in a front-rear direction of the vehicle toward a lower side.

2. The vehicle upper structure according to claim 1, wherein a lower end portion of the rear side surface of the wall is inclined rearward in the front-rear direction of the vehicle toward a lower side.

3. The vehicle upper structure according to claim 2, wherein a position of the lower end portion of the rear side surface of the wall in the front-rear direction of the vehicle is at the same position as a position of the upper end portion of the rear side surface of the wall in the front-rear direction of the vehicle, or is at a rear side in the front-rear direction of the vehicle than the position of the upper end portion of the rear side surface of the wall in the front-rear direction of the vehicle.

4. The vehicle upper structure according to claim 2, wherein the rear side surface of the wall is curved along a trajectory of movement of the front end of the hatchback door due to opening or closing of the hatchback door.

5. The vehicle upper structure according to claim 1, wherein the rail garnish is made of a resin.